Experiment No.9

Implementing Named Entity Recognition (NER) using NLTK , SpaCy and Stanford NER in Python AND using INLTK for regional language.

NLTK

```
import nltk
# Download required NLTK resources
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('averaged_perceptron_tagger_eng')
nltk.download('maxent_ne_chunker')
nltk.download('maxent_ne_chunker_tab')
nltk.download('words')
# Input sentence
text = """
On 26 January 2027 at 9:00 AM, Priya Mehta from Google purchased
3 Samsung Galaxy S26 phones in New Delhi, India, for ₹1,80,000
with a 15% discount during the Republic Day celebration.
"""
# Tokenization
tokens = nltk.word_tokenize(text)

# POS Tagging
pos_tags = nltk.pos_tag(tokens)

# Named Entity Recognition
named_entities = nltk.ne_chunk(pos_tags)

# Display complete NER tree
print("Named Entity Recognition using NLTK:\n")
print(named_entities)

# Extract and display classified entities
print("\nClassified Entities:\n")

for chunk in named_entities:
    if hasattr(chunk, 'label'):
        entity = " ".join(word for word, tag in chunk)
        print(f"{entity} -> {chunk.label_ if hasattr(chunk, 'label_') else chunk.label()}")
```

Output
Classified Entities:

Priya Mehta -> PERSON
Google -> PERSON
Samsung Galaxy S26 -> ORGANIZATION
New Delhi -> GPE
India -> GPE
Republic -> ORGANIZATION

SpaCy

```
import spacy

# Load the English spaCy model
nlp = spacy.load("en_core_web_sm")

# Input sentence
text = """
On 26 January 2027 at 9:00 AM, Priya Mehta from Google purchased
3 Samsung Galaxy S26 phones in New Delhi, India, for ₹1,80,000
with a 15% discount during the Republic Day celebration.
"""

# Process the text
doc = nlp(text)

# Display named entities
print("Named Entity Recognition using spaCy:\n")

for ent in doc.ents:
    print(f"{ent.text} -> {ent.label_} -> {spacy.explain(ent.label_)}")
```

Output

```
Named Entity Recognition using spaCy:

26 January 2027 -> DATE -> Absolute or relative dates or periods
9:00 AM -> TIME -> Times smaller than a day
Priya Mehta -> PERSON -> People, including fictional
Google -> ORG -> Companies, agencies, institutions, etc.
3 -> CARDINAL -> Numerals that do not fall under another type
Samsung Galaxy S26 -> ORG -> Companies, agencies, institutions, etc.
New Delhi -> GPE -> Countries, cities, states
India -> GPE -> Countries, cities, states
1,80,000 -> MONEY -> Monetary values, including unit
15% -> PERCENT -> Percentage, including "%"
```

Stanza

```
# Install Stanza
!pip install -q stanza

import stanza

# Download English model
stanza.download("en")

# Create NLP pipeline
nlp = stanza.Pipeline(
    lang="en",
    processors="tokenize,ner"
)
```

```
# Input sentence
text = """
On 26 January 2027 at 9:00 AM, Priya Mehta from Google purchased
3 Samsung Galaxy S26 phones in New Delhi, India, for ₹1,80,000
with a 15% discount during the Republic Day celebration.
"""

# Process text
doc = nlp(text)

print("Named Entity Recognition using Stanza:\n")
```

Output

```
Named Entity Recognition using Stanza:

26 January 2027 -> DATE
9:00 AM -> TIME
Priya Mehta -> PERSON
Google -> ORG
3 -> CARDINAL
Samsung Galaxy S26 -> PRODUCT
New Delhi -> GPE
India -> GPE
₹1,80,000 -> MONEY
15% -> PERCENT
Republic Day -> EVENT
```

Hindi

```
import stanza

# Create Hindi NLP pipeline
nlp = stanza.Pipeline(
   lang="hi",
   processors="tokenize,ner"
)

# Hindi input sentence
text = """
26 जनवरी 2027 को सुबह 9 बजे, प्रिय शर्मा ने दिल्ली में
Google से ₹1,80,000 में 3 Samsung Galaxy S26 फोन खरीदे।
"""

# Process the text
doc = nlp(text)

print("Named Entity Recognition using Stanza:\n")

# Print named entities
for ent in doc.ents:
```

```
    print(f"{ent.text} -> {ent.type}")
```

Output

```
26 जनवरी 2027 -> NETI
9 -> NEN
प्रिय शर्मा -> NEP
दिल्ली -> NEL
₹1,80,000 -> NEN
3 -> NEN
```

Marathi

```
import stanza

# Create Marathi NLP pipeline
nlp = stanza.Pipeline(
    lang="mr",
    processors="tokenize,ner"
)

# Marathi sentence
text = """
26 जानेवारी 2027 रोजी सकाळी 9 वाजता, प्रिया शर्मा यांनी
मुंबई, महाराष्ट्र येथे Google कडून ₹1,80,000 मध्ये
Samsung Galaxy S26 चे 3 फोन खरेदी केले.
"""

# Process the text
doc = nlp(text)

print("Named Entity Recognition using Stanza:\n")

# Print entities
for ent in doc.ents:
    print(f"{ent.text} -> {ent.type}")
```

Output

```
Named Entity Recognition using Stanza:

26 जानेवारी -> NED
202 -> NEM
7 -> NEM
सकाळी -> NETI
9 -> NETI
प्रिया शर्मा -> NEP
मुंबई -> NEL
महाराष्ट्र -> NEL
₹1 -> NEM
80 -> NEM
```

000 -> NEM
3 -> NEM